

Linear Models Lecture 12: Hypothesis Testing

Robert Gulotty

University of Chicago

April 29, 2026

Where We Are

Last lecture: We derived the Wald statistic

$$W = (R\hat{\beta} - r)' [R\hat{V}R']^{-1} (R\hat{\beta} - r) \xrightarrow{d} \chi_q^2$$

and showed how to construct robust tests and confidence intervals.

Today: We go deeper into the *practice* of hypothesis testing.

- Constrained least squares: equality *and* inequality constraints
- Partial identification: when the data only pins β to a feasible set
- The F test (recap), Score (LM) tests, and the trinity of classical tests
- Test inversion for confidence regions
- Multiple testing and Bonferroni corrections
- Power: what determines whether you can detect an effect?
- Hansen's practical advice for applied work

Constrained Least Squares: Setup

Many tests today compare an **unrestricted** model to a **restricted** one. We need to know how to estimate under restrictions.

Problem: Minimize the sum of squared errors subject to q linear constraints:

$$\tilde{\beta}_{\text{CLS}} = \arg \min_{\beta} \sum_{i=1}^n (Y_i - \mathbf{X}_i' \beta)^2 \quad \text{subject to} \quad \mathbf{R}' \beta = \mathbf{r}$$

where $\mathbf{R}$ is $k \times q$ and $\mathbf{r}$ is $q \times 1$.

Examples:

- $\beta_3 = 0$: set $\mathbf{R}' = (0, 0, 1, 0, \dots)$, $\mathbf{r} = 0$ (exclusion restriction)
- $\beta_2 = \beta_3$: set $\mathbf{R}' = (0, 1, -1, 0, \dots)$, $\mathbf{r} = 0$ (equality restriction)
- $\beta_2 + \beta_3 = 1$: set $\mathbf{R}' = (0, 1, 1, 0, \dots)$, $\mathbf{r} = 1$ (adding-up constraint)

The CLS Estimator

Solve via Lagrange multipliers. The solution is:

$$\tilde{\beta}_{\text{CLS}} = \hat{\beta}_{\text{OLS}} - (\mathbf{X}'\mathbf{X})^{-1}\mathbf{R} [\mathbf{R}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}]^{-1} (\mathbf{R}'\hat{\beta}_{\text{OLS}} - \mathbf{r})$$

Reading the formula:

- Start from unrestricted OLS $\hat{\beta}_{\text{OLS}}$, subtract a correction toward the constraint
- Correction is proportional to $(\mathbf{R}'\hat{\beta}_{\text{OLS}} - \mathbf{r})$: how far OLS violates H_0
- If OLS already satisfies the restriction, $\tilde{\beta}_{\text{CLS}} = \hat{\beta}_{\text{OLS}}$

The restricted residuals and variance estimate are:

$$\tilde{e}_i = Y_i - \mathbf{x}_i'\tilde{\beta}_{\text{CLS}}, \quad \tilde{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n \tilde{e}_i^2$$

Since the restriction constrains the parameter space: $\tilde{\sigma}^2 \geq \hat{\sigma}^2$ always.

CLS in Practice

Simple case: If H_0 sets some coefficients to zero, CLS just means dropping those regressors.

```
# Unrestricted
mod_U <- lm(lwage ~ education + exper + I(exper^2) +
            female + union, data = wages)
# Restricted (H0: beta_female = beta_union = 0)
mod_R <- lm(lwage ~ education + exper + I(exper^2),
            data = wages)
```

General case: For arbitrary linear restrictions (e.g., $\beta_2 = \beta_3$), reparameterize or use the closed-form formula.

The key outputs from CLS are $SSE_R = \sum \tilde{e}_i^2$ and $\tilde{\sigma}^2$, which we use to build the F and Score statistics.

Worked Example: CLS for a Linear Restriction

Setup. Wage regression with separate returns to general experience and firm-specific tenure:

$$\log w_i = \beta_0 + \beta_1 \text{educ}_i + \beta_2 \text{exper}_i + \beta_3 \text{tenure}_i + e_i.$$

Becker–Mincer human-capital theory predicts these have *different* returns: firm-specific human capital should pay more at the current employer than general experience does. We want to *test* this — formally, $H_0 : \beta_2 = \beta_3$ vs. $H_1 : \beta_2 \neq \beta_3$.

Important framing. We do not impose the restriction because we believe it. We compute the restricted estimator to obtain SSE_R , which feeds the F statistic that *tests* the restriction.

Approach 1: Reparameterize. Under H_0 the model becomes

$$\log w_i = \beta_0 + \beta_1 \text{educ}_i + \beta_2 (\text{exper}_i + \text{tenure}_i) + e_i,$$

which is just OLS on the new combined regressor:

```
mod_R <- lm(lwage ~ educ + I(exper + tenure), data = wages)
```

Beyond Equality: Inequality Constraints

The CLS formulation extends naturally to inequalities:

$$\tilde{\beta}_{\text{CLS}} = \arg \min_{\beta \in \mathcal{B}} \sum_{i=1}^n (Y_i - \mathbf{x}_i' \beta)^2$$

where the **feasible set** $\mathcal{B} \subseteq \mathbb{R}^k$ is defined by linear inequalities $\mathbf{R}'\beta \leq \mathbf{r}$.

Examples:

- **Sign restrictions:** $\beta_j \geq 0$ (a variance, a price-elasticity sign, a share)
- **Box constraints:** $0 \leq \beta_j \leq 1$ (mixing weights, probabilities)
- **Monotonicity:** $\beta_1 \leq \beta_2 \leq \beta_3$ (ordered effects)
- **Theory bounds:** $\beta \in [\underline{\beta}, \overline{\beta}]$ from prior knowledge

The feasible set $\mathcal{B}$ is a *polyhedron* (an intersection of half-spaces) rather than a single point or a linear subspace.

Solving Inequality CLS: Two Cases

Strategy. Solve unconstrained OLS first, then check whether the result satisfies the constraints.

Case 1: $\hat{\beta}_{OLS} \in \mathcal{B}$. No constraint binds; the constrained and unconstrained problems have the same solution:

$$\tilde{\beta}_{CLS} = \hat{\beta}_{OLS}.$$

Case 2: $\hat{\beta}_{OLS} \notin \mathcal{B}$. The constrained solution sits on the boundary of $\mathcal{B}$. On the face where one or more constraints bind, those constraints hold with equality, and the solution is given by the *equality*-CLS formula from earlier in this lecture, applied to the binding constraints only.

Conceptually, you are projecting $\hat{\beta}_{OLS}$ onto the feasible region $\mathcal{B}$. Picking which constraints bind is what makes the inequality case harder than the equality case — in practice this is solved by a *quadratic program*. In R: `quadprog::solve.QP()`.

Why Inequality Constraints? Partial Identification

Sometimes the data and minimal assumptions only tell us β lies in a *set*, not a point.

The identified set: $\Theta_I = \{\beta : \beta \text{ is consistent with the data and assumptions}\}$.

- **Point identification:** $\Theta_I = \{\beta_0\}$ — a single value
- **Partial identification:** Θ_I is a non-degenerate set, often a polyhedron

Where it shows up: missing data, censoring, interval-valued outcomes, treatment effect bounds, set-identified moment inequalities.

The identified set *is* a feasible set $\mathcal{B}$. Estimation under partial identification reports both a point (e.g., the inequality-CLS solution on $\mathcal{B}$) and the *width* of $\mathcal{B}$ — and the width does *not* shrink with n .

A Worked Example: NSW Job Training

The **National Supported Work Demonstration** (1976–77) randomly assigned disadvantaged workers to a 9–12 month training program ($X_i = 1$) or a control group ($X_i = 0$). LaLonde (1986) used it to benchmark non-experimental evaluators.

Outcome. $Y_i = 1$ if subject i had positive earnings in 1978 (employed at follow-up).

Sample (Dehejia–Wahba experimental sample, $n = 445$):

	n	Employed in 1978	$\bar{Y}_{\text{obs}}$
Treatment ($X = 1$)	185	140	0.757
Control ($X = 0$)	260	168	0.646

Naive ATE: $\bar{Y}_{\text{obs}}(1) - \bar{Y}_{\text{obs}}(0) = 0.757 - 0.646 = 0.111$ (an 11 pp employment effect).

NSW's actual follow-up was nearly complete. **What if 30% had been lost?** The next slides treat that as a thought experiment to see what the data could still pin down.

The Manski Decomposition (30% Missing)

Suppose only 70% of subjects could be located in 1978. Let $D_i = 1$ if observed; $P(D_i = 1) = 0.7$.

For each arm, by the law of total probability:

$$E[Y \mid X = x] = \underbrace{p(x) \bar{Y}_{\text{obs}}(x)}_{\text{identified}} + \underbrace{(1 - p(x)) E[Y \mid X = x, D = 0]}_{\text{unknown}}$$

The first term is identified from observed respondents. The second is unknown — but since $Y \in \{0, 1\}$, the unknown conditional mean lies in $[0, 1]$, so:

$$p(x) \bar{Y}_{\text{obs}}(x) \leq E[Y \mid X = x] \leq p(x) \bar{Y}_{\text{obs}}(x) + (1 - p(x))$$

The lower bound replaces every missing Y with 0 (worst case for that arm); the upper bound replaces every missing Y with 1. These are **sharp**: no narrower bound is consistent with the data alone.

Worst-Case Bounds: Apply to NSW

Plug in $p(x) = 0.7$ for each arm.

Treatment ($\bar{Y}_{\text{obs}} = 0.757$):

$$0.7 \cdot 0.757 \leq E[Y \mid X=1] \leq 0.7 \cdot 0.757 + 0.3 \implies E[Y \mid X=1] \in [0.530, 0.830]$$

Control ($\bar{Y}_{\text{obs}} = 0.646$):

$$0.7 \cdot 0.646 \leq E[Y \mid X=0] \leq 0.7 \cdot 0.646 + 0.3 \implies E[Y \mid X=0] \in [0.452, 0.752]$$

ATE bounds. $\theta = E[Y \mid X=1] - E[Y \mid X=0]$ is smallest when arm 1 is at its lower bound *and* arm 0 at its upper bound (largest in the opposite corner):

$$\theta \in [0.530 - 0.752, 0.830 - 0.452] = [-0.22, 0.38]$$

The naive estimate 0.11 sits inside, but the bound is wide enough that we cannot even sign the effect. Width = 0.60.

Tighter Bounds via Monotone Selection

Plausible assumption. Subjects lost to follow-up — transient, moving, out of the labor force — are weakly *less* likely to be employed than those we observe in the same arm:

$$E[Y \mid X = x, D = 0] \leq E[Y \mid X = x, D = 1] = \bar{Y}_{\text{obs}}(x).$$

Effect. The upper bound now replaces missing Y with $\bar{Y}_{\text{obs}}(x)$, not with 1, so $E[Y \mid X=x] \leq \bar{Y}_{\text{obs}}(x)$. The lower bound is unchanged.

Apply to NSW:

$$E[Y \mid X=1] \in [0.530, 0.757], \quad E[Y \mid X=0] \in [0.452, 0.646],$$

$$\theta \in [0.530 - 0.646, 0.757 - 0.452] = [-0.12, 0.30].$$

Width drops from 0.60 to 0.42. Still crosses zero — an 11 pp underlying effect is too small for one mild assumption to settle the sign at 30% missing.

The Identified Set as an Inequality-CLS Feasible Set

Write the CEF as a linear regression $E[Y | X] = \beta_0 + \beta_1 X$, so $\beta_0 = E[Y | X=0]$ and $\beta_0 + \beta_1 = E[Y | X=1]$.

The four MTS bounds map directly to four linear inequalities on $\beta = (\beta_0, \beta_1)'$:

$$\mathcal{B} = \{\beta : 0.452 \leq \beta_0 \leq 0.646, 0.530 \leq \beta_0 + \beta_1 \leq 0.757\}.$$

Inequality CLS on the observed sample:

$$\tilde{\beta}_{\text{CLS}} = \arg \min_{\beta \in \mathcal{B}} \sum_{i: D_i=1} (Y_i - \beta_0 - \beta_1 X_i)^2.$$

$\mathcal{B}$ is the identified set: it pins β_1 to the interval $[-0.12, 0.30]$. An inequality-CLS point estimate is just one β inside $\mathcal{B}$; reporting that point hides the bound. Report the interval.

Computing the Bounds in R

```
u <- "https://users.nber.org/~rdehejia/data/nswre74_"
cn <- c("treat", "age", "educ", "black", "hispan", "married",
        "nodegree", "re74", "re75", "re78")
d <- rbind(read.table(paste0(u, "treated.txt"), col.names=cn),
            read.table(paste0(u, "control.txt"), col.names=cn))
y1 <- mean(d$re78[d$treat==1] > 0)      # 0.7568      (n = 185)
y0 <- mean(d$re78[d$treat==0] > 0)      # 0.6462      (n = 260)
p <- 0.7                                # 70% located, 30% missing
ate_wc <- c(p*y1 - (p*y0+1-p), (p*y1+1-p) - p*y0) # [-0.22, 0.38]
ate_mts <- c(p*y1 - y0, y1 - p*y0)          # [-0.12, 0.30]
```

What to report: “The NSW employment ATE lies in $[-0.12, 0.30]$ under monotone selection (naive estimate 0.11).”

Setting Up the Linear Program

Reparameterize from β to the conditional means $\mu = (\mu_0, \mu_1)'$ where $\mu_x = E[Y | X=x]$. Then $\beta_0 = \mu_0$ and $\beta_1 = \mu_1 - \mu_0 = \mathbf{c}'\mu$ with $\mathbf{c} = (-1, 1)'$.

Each MTS bound is a row $\mathbf{a}_j'\mu \{ \geq, \leq \} r_j$. Stack the four:

$$\underbrace{\begin{pmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{pmatrix}}_A \underbrace{\begin{pmatrix} \mu_0 \\ \mu_1 \end{pmatrix}}_{\text{dir}} \underbrace{\begin{matrix} \geq \\ \leq \\ \geq \\ \leq \end{matrix}}_{\text{dir}} \underbrace{\begin{pmatrix} 0.452 \\ 0.646 \\ 0.530 \\ 0.757 \end{pmatrix}}_{\text{rhs}}.$$

Why μ and not β ? lpSolve defaults variables to ≥ 0 ; conditional means satisfy this automatically, but β_1 can be negative.

Bounds via lpSolve

Two LPs — minimize and maximize $\mathbf{c}'\boldsymbol{\mu} = \mu_1 - \mu_0$ over the same feasible set:

```
library(lpSolve)
A      <- rbind(c(1,0), c(1,0), c(0,1), c(0,1))
dir    <- c(">=", "<=", ">=", "<=")
rhs    <- c(0.452, 0.646, 0.530, 0.757)
c(b1_lo = lp("min", c(-1,1), A, dir, rhs)$objval,
  b1_hi = lp("max", c(-1,1), A, dir, rhs)$objval)
# b1_lo  b1_hi:  -0.116  0.305
```

lpSolve confirms $\beta_1 \in [-0.12, 0.30]$. More parameters or constraints just adds rows to A , dir , and rhs .

The F Test (Recap from Section)

Compare the unrestricted SSE to the restricted (CLS) SSE. Testing $H_0: \mathbf{R}'\boldsymbol{\beta} = \mathbf{r}$ with q restrictions:

$$F = \frac{(\text{SSE}_R - \text{SSE}_U)/q}{\text{SSE}_U/(n-k)} = \frac{W^0}{q}$$

- **Distribution:** under normal homoskedastic errors, $F \sim F_{q, n-k}$ exactly; asymptotically (any iid errors), $qF \xrightarrow{d} \chi_q^2$.
- **Heteroskedasticity:** the $F_{q, n-k}$ distribution is no longer valid — use the robust Wald test from Lecture 11.
- **In R:** `anova(mod_R, mod_U)`.

The package-default “F-statistic that all slopes equal zero” is rarely a scientifically interesting hypothesis. Hansen: there is no reason to report it.

Score Tests: The Idea

The Wald test starts from the **unrestricted** estimate and asks: is $\hat{\theta}$ far from θ_0 ?

The **Score test** (also called **Lagrange Multiplier test**) starts from the **restricted** estimate and asks: does the objective function want to move away from the restriction?

- Compute the restricted estimator $\tilde{\beta}$ (constrained least squares)
- Evaluate the *gradient* (score) of the objective function at $\tilde{\beta}$
- If H_0 is true, the score should be near zero (we're near the optimum)
- If H_0 is false, the score should be large (the restriction is pulling us away)

Score Statistic for Linear Restrictions

For $H_0: \mathbf{R}'\boldsymbol{\beta} = \mathbf{c}$ in the normal regression model:

$$S = \frac{(\mathbf{R}'\hat{\boldsymbol{\beta}} - \mathbf{c})'[\mathbf{R}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}]^{-1}(\mathbf{R}'\hat{\boldsymbol{\beta}} - \mathbf{c})}{\tilde{\sigma}^2}$$

- Identical to W^0 except s^2 replaced by $\tilde{\sigma}^2$ (restricted variance).
- Under H_0 : $S \xrightarrow{d} \chi_q^2$

Key result: S is a monotone transformation of F : $S = n \left(1 - \frac{1}{1 + qF/(n-k)} \right)$, so the Score test and the F test always give the same accept/reject decision.

When Are Score Tests Useful?

For linear regression with linear restrictions, the Score test offers nothing new over F or Wald.

But in more complex settings, Score tests have a major advantage:

- They only require estimation under H_0 (the restricted model)
- The “unrestricted” model may not even be a simple OLS regression

Example: Testing for heteroskedasticity (Breusch–Pagan). The null is homoskedastic OLS—easy. The unrestricted model allows $\text{Var}(e_i | X_i)$ to vary with X , which requires modeling the variance function. The Score test only needs the OLS residuals.

Same idea applies to:

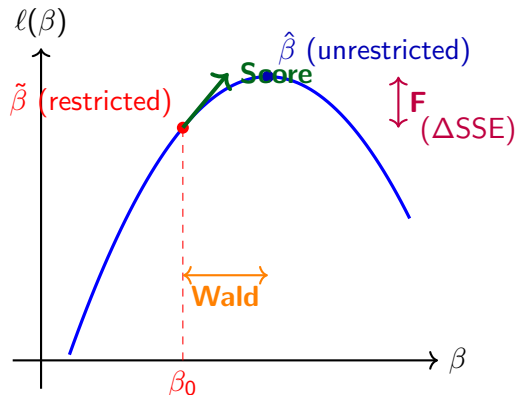
- Testing for serial correlation (Breusch–Godfrey): null is OLS with iid errors
- Testing for omitted nonlinearities: null is a simple linear specification
- Any setting where H_0 gives you a clean model but H_1 is messy

Three Ways to Test the Same Hypothesis

	Wald	Score (LM)	F / LR-like
Estimates from Measures	Unrestricted Distance of $\hat{\theta}$ from θ_0	Restricted Gradient at restriction	Both Change in fit (SSE)
Null dist.	χ_q^2	χ_q^2	χ_q^2/q or $F_{q,n-k}$
Robust?	Yes (sandwich)	Score-like	No

For linear restrictions in the normal model, all three are monotone transformations of each other and give identical decisions. They differ for nonlinear restrictions or non-normal models.

Visualizing the Trinity



- **Wald**: horizontal distance between $\hat{\beta}$ and β_0
- **Score**: slope at restricted estimate
- **F**: vertical distance (ΔSSE)

Confidence Intervals Are Inverted Tests

Recall the standard 95% CI:

$$\hat{C} = \left[\hat{\theta} - 1.96 \cdot s(\hat{\theta}), \hat{\theta} + 1.96 \cdot s(\hat{\theta}) \right]$$

This is exactly the set of values θ that are *not rejected* by a two-sided t -test:

$$\hat{C} = \{ \theta : |T(\theta)| \leq 1.96 \}$$

General principle: Inverting a test with good Type I error control produces a confidence set with good coverage.

$$P[\theta \in \hat{C}] = P[\text{Accept} \mid \theta] = 1 - P[\text{Type I error}]$$

Why Test Inversion Matters: Nonlinear Parameters

Consider $\theta = \beta_1/\beta_2$ (e.g., peak experience in a wage equation).

Approach 1: Delta method CI

$$\hat{\theta} \pm 1.96 \cdot s(\hat{\theta})$$

From Lecture 11, this can be inaccurate for ratios (Fieller's problem).

Approach 2: Rewrite as a *linear* restriction $\beta_1 - \theta\beta_2 = 0$ and invert:

$$\hat{C} = \left\{ \theta : \frac{(\hat{\beta}_1 - \theta\hat{\beta}_2)^2}{\mathbf{R}'(\theta)\hat{\mathbf{V}}\mathbf{R}(\theta)} \leq 1.96^2 \right\}$$

where $\mathbf{R}(\theta) = (1, -\theta)'$. This requires a grid search over θ .

Hansen's example: Peak experience = $-50\beta_1/\beta_2$.

Delta method CI: [29.8, 29.9]. Inverted linear test CI: [29.1, 30.6].

The delta method interval is **far too narrow**.

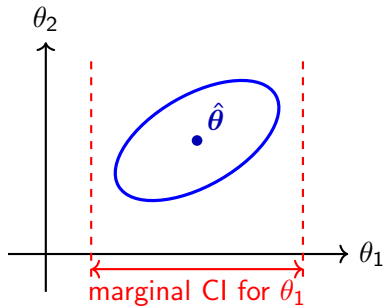
Test Inversion in R

```
# Wage equation: log(wage) ~ exper + exper^2/100 + ...
mod <- lm(lwage ~ exper + I(exper^2/100) + education,
          data = wages)
b <- coef(mod); V <- vcovHC(mod)

# Grid search: invert the t-test for theta = -50*b1/b2
theta_grid <- seq(20, 40, by = 0.01)
in_CI <- sapply(theta_grid, function(th) {
  R <- c(0, 1, -th/50, 0) # gradient of b[2] - th*b[3]/50
  num <- (b[2] - th * b[3] / 50)^2
  den <- t(R) %*% V %*% R
  num / den <= qchisq(0.95, 1)
})
CI <- range(theta_grid[in_CI])
```

Confidence Regions for Multiple Parameters

For $q > 1$ parameters, invert the Wald test: $\hat{C} = \{\boldsymbol{\theta} : W(\boldsymbol{\theta}) \leq \chi_{q, 1-\alpha}^2\}$ — an **ellipsoid** in $\mathbb{R}^q$ centered at $\hat{\boldsymbol{\theta}}$, shaped by $\hat{\mathbf{V}}_{\boldsymbol{\theta}}^{-1}$.



Marginal CIs (projections) are always wider than the ellipsoid in each dimension.

The Multiple Testing Problem

Suppose you test k hypotheses, each at level $\alpha = 0.05$. Under the **global null** (all k true), what is $P[\text{at least one false rejection}]$?

By **Boole's inequality**:

$$P\left[\min_{j \leq k} p_j < \alpha\right] \leq \sum_{j=1}^k P[p_j < \alpha] \rightarrow k\alpha$$

k (tests)	α per test	Familywise error $\leq$
5	0.05	0.25
10	0.05	0.50
20	0.05	1.00

With 20 tests, you are *virtually certain* to get a false rejection!

The Bonferroni Correction

Goal: Control the *familywise error rate* (FWER) at α .

Rule: Reject the j th hypothesis only if $p_j < \alpha/k$. Equivalently: $p_{\text{Bonf}} = k \cdot \min_{j \leq k} p_j$.

Proof:

$$P\left[\min_{j \leq k} p_j < \frac{\alpha}{k}\right] \leq \sum_{j=1}^k P\left[p_j < \frac{\alpha}{k}\right] \rightarrow k \cdot \frac{\alpha}{k} = \alpha$$

Simple and conservative. Controls FWER for *any* dependence structure among the tests. The cost: reduced power when k is large.

Bonferroni: Worked Example

Two coefficients tested at $\alpha = 0.05$:

	Individual p	Bonferroni $p (= 2 \times p)$
Union membership	0.04	0.08
Married status	0.15	0.30

- **Without correction:** Union membership “significant” at 5%.
- **With Bonferroni:** Neither significant at $\text{FWER} = 0.05$ (need $p < 0.025$).

When to worry: exploring many specifications/subgroups, examining many coefficients, or reporting the “most significant” result.

Bonferroni in R

```
# Get p-values from a regression
mod <- lm(lwage ~ education + exper + I(exper^2) +
          female + union + married, data = wages)
pvals <- summary(mod)$coefficients[-1, 4] # drop intercept

# Bonferroni correction
p_bonf <- p.adjust(pvals, method = "bonferroni")
cbind(raw = round(pvals, 4), bonferroni = round(p_bonf, 4))

# Other options: Holm (less conservative, still controls FWER)
p_holm <- p.adjust(pvals, method = "holm")
```

Holm's method is uniformly more powerful than Bonferroni while still controlling FWER. Use it when available.

Type I and Type II Errors

	H_0 true	H_0 false
Reject H_0	Type I error (α)	Correct (Power = $1 - \beta$)
Accept H_0	Correct ($1 - \alpha$)	Type II error (β)

- **Size** (α): probability of falsely rejecting a true null
- **Power** (π): probability of correctly rejecting a false null: $\pi(\theta) = P[\text{Reject } H_0 \mid \theta \neq \theta_0]$
- Power depends on the *true value* of θ , on n , and on α .

Key trade-off: Making α smaller reduces Type I errors but also reduces power. A test with $\alpha = 0$ never rejects and has zero power.

Power of the t -Test

For $H_0: \theta = \theta_0$ vs. $H_1: \theta > \theta_0$:

$$T = \frac{\hat{\theta} - \theta_0}{s(\hat{\theta})} \approx Z + \delta, \quad Z \sim N(0, 1)$$

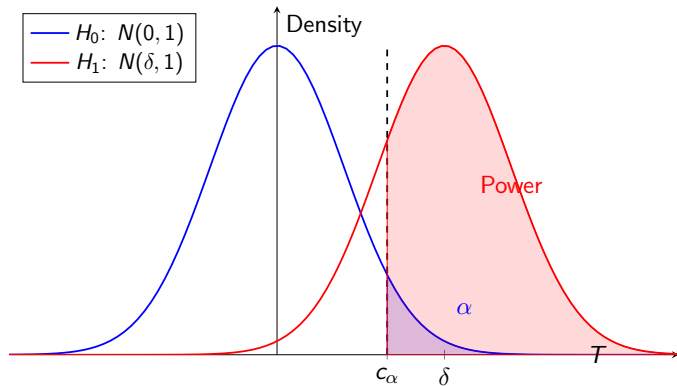
where $\delta = (\theta - \theta_0)/s(\hat{\theta})$ is the **signal-to-noise ratio**.

Power of a one-sided test at level α : $\pi(\delta) = \Phi(\delta - z_\alpha)$

Power increases when:

- The true effect $|\theta - \theta_0|$ is larger
- The standard error $s(\hat{\theta})$ is smaller (more precise estimation)
- The significance level α is larger (less stringent threshold)
- The sample size n is larger (since $s(\hat{\theta}) \propto 1/\sqrt{n}$)

Visualizing Power



Power in OLS: What You Can Control

In OLS, $s(\hat{\theta}) \approx \sigma_e / (\text{sd}(X) \cdot \sqrt{n})$ for a single regressor. So:

$$\delta \approx \frac{(\theta - \theta_0) \cdot \text{sd}(X) \cdot \sqrt{n}}{\sigma_e}$$

Levers for increasing power:

- 1 **Increase** n : power grows with $\sqrt{n}$
- 2 **Reduce** σ_e : add controls that explain Y (reduce residual variance)
- 3 **Increase** $\text{sd}(X)$: more variation in the regressor of interest
- 4 **Increase** α : use 5% instead of 1% (but more Type I errors)

Controls help power! Relevant covariates reduce σ_e without reducing $\text{sd}(X)$ —a free power boost. But irrelevant covariates cost degrees of freedom.

Power and Test Dimension

For joint tests ($q > 1$), the Wald statistic under local alternatives: $W \xrightarrow{d} \chi_q^2(\lambda)$, where $\lambda = \mathbf{h}' \mathbf{V}_\theta^{-1} \mathbf{h}$ is the **non-centrality parameter**.

Critical fact: For a fixed λ , power *decreases* as q increases.

q (restrictions)	λ for 50% power	Sample size increase
1	3.85	baseline
2	4.96	+28%
3	5.77	+50%

Takeaway: Testing more restrictions simultaneously dilutes power. A single-coefficient t -test is more powerful than a joint F -test that includes other restrictions.

The 50% Power Benchmark

How far must the true parameter be from θ_0 for 50% power?

One-sided test:

- At $\alpha = 0.05$: need $\delta \geq 1.65$ standard errors
- At $\alpha = 0.01$: need $\delta \geq 2.33$ standard errors

Implication for sample size: $(2.33/1.65)^2 \approx 2$.

A test at $\alpha = 0.01$ requires roughly **twice the sample size** as $\alpha = 0.05$ to achieve the same power.

Two-sided ($\alpha = 0.05$): need $|\delta| \geq 1.96$ for 50% power. Since $se \propto 1/\sqrt{n}$, you need $n \propto (\sigma_e/(\theta - \theta_0))^2$.

Power Calculation in R

```
# Approximate power for a two-sided t-test in OLS
# Inputs: effect size, residual SD, regressor SD, n, alpha
power_ols <- function(effect, sigma_e, sd_x, n, alpha=0.05) {
  se <- sigma_e / (sd_x * sqrt(n))
  delta <- effect / se
  z <- qnorm(1 - alpha/2)
  power <- pnorm(delta - z) + pnorm(-delta - z)
  return(power)
}

# Example: detect beta = 0.1 with sigma_e = 1, sd(x) = 2
sapply(c(50, 100, 200, 500, 1000), function(n)
  round(power_ols(0.1, 1, 2, n), 3))
# [1] 0.080 0.117 0.198 0.463 0.803
```

With $\beta = 0.1$, $\sigma_e = 1$, $\text{sd}(X) = 2$: you need $n \approx 1000$ for 80% power!

Test Consistency

Definition

A test is **consistent against fixed alternatives** if for any true $\theta \neq \theta_0$: $P[\text{Reject } H_0 \mid \theta] \rightarrow 1$ as $n \rightarrow \infty$.

Good news: The t -test and Wald test are consistent. As $n \rightarrow \infty$, $s(\hat{\theta}) \rightarrow 0$, so $|T| \rightarrow \infty$ whenever $\theta \neq \theta_0$.

Caution: Consistency means you will eventually reject any false null—including *economically trivial* deviations from θ_0 .

In very large samples, statistical significance $\neq$ economic significance. A statistically significant but tiny coefficient may not be meaningful.

Hansen's Rules for Applied Work

1. Report standard errors, not t -ratios.

- Standard errors focus attention on precision and confidence intervals.

2. Report p -values, not asterisks.

- p -values contain more information than $*/**/* ** *$ (“an inferior practice”).

3. Focus on economically motivated hypotheses.

- Don't mechanically test every coefficient against zero.
- Report the t -test for $\beta_j = 0$ when this is a scientifically interesting question.

More Practical Advice

4. “Do Not Reject” $\neq$ “Accept.”

- Failing to reject means insufficient evidence, *not* that H_0 is true.
- Never write: “the regression finds that X has no effect on Y .”

5. Statistical significance $\neq$ economic significance.

- With large n , even tiny effects become “significant.”
- Always discuss the *magnitude* and substantive meaning.

6. For nonlinear hypotheses, use minimum distance or test inversion.

- The Wald statistic is *not invariant* to the algebraic formulation of H_0 .
- Different equivalent formulations can give different results!

The “What to Report” Checklist

For any coefficient of interest θ :

- 1 **Point estimate** $\hat{\theta}$: the “best guess”
- 2 **Standard error** $s(\hat{\theta})$: measure of precision
- 3 **Confidence interval**: range of values consistent with the data
- 4 **Economic interpretation**: what the magnitude means in context

When testing:

- 5 State the **hypothesis** clearly (what question does this answer?)
- 6 Report the ***p*-value** (not just “significant” / “not significant”)
- 7 Discuss **power** if you fail to reject (could you have detected a meaningful effect?)
- 8 Account for **multiple testing** if examining many hypotheses

Summary

Topic	Key Takeaway
CLS	Equality $R'\beta = r$ via Lagrange; gives SSE_R for F and Score
Inequality CLS	Feasible set $\mathcal{B}$ as a polyhedron; project OLS onto $\mathcal{B}$ via QP
Partial ID	Θ_I has width that does not vanish with n ; report bounds, not just a point
F test	$F = W^0/q; \sim F_{q,n-k}$ under normal homo.; use robust Wald otherwise
Score test	From restricted model; equivalent to F for linear restrictions
Trinity	Wald, Score, F agree for linear H_0 in normal model
Test inversion	Confidence sets = non-rejected values; essential for nonlinear parameters
Bonferroni	With k tests, use α/k to control familywise error
Power	δ = effect/se; grows with $\sqrt{n}$; 50% power needs $\delta \approx 2$
Practical advice	Report SEs, CIs, p -values; focus on magnitudes; consider power

Next Time

Lecture 13: Instrumental Variables

- What happens when $E[\mathbf{X}\mathbf{e}] \neq \mathbf{0}$?
- Omitted variable bias, simultaneity, measurement error
- The IV solution: find $\mathbf{Z}$ such that $E[\mathbf{Z}\mathbf{e}] = \mathbf{0}$ but $E[\mathbf{Z}\mathbf{X}'] \neq \mathbf{0}$